# BREAST CANCER CLASSIFICATION ANALYSIS REPORT

Programming For Data Analytics 1

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## Contents

# Predictive Modelling for Cancer Diagnosis Support Using Supervised Machine Learning

## Introduction

A medical scheme requested a machine learning-based solution to improve and speed up the approval process for cancer-related medical benefits. This was motivated by the need for faster identification of high-risk patients and improved decision-making in healthcare operations.

This project applied supervised machine learning techniques to classify breast tumours as either benign or malignant using the Breast Cancer Wisconsin Dataset. The aim was to evaluate how predictive modelling can support faster and more reliable medical decisions in a healthcare environment (Müller and Guido, 2016).

The dataset was selected due to its structured numerical features and suitability for binary classification problems commonly used in medical machine learning research (Springer, 2017).

## Methodology

The project followed a standard supervised learning pipeline grounded in established machine learning frameworks (Müller and Guido, 2016). The process began with importing the dataset and required Python libraries for data processing, visualisation, and modelling.

Exploratory Data Analysis (EDA) was conducted to understand data structure, detect anomalies, and assess feature distributions. This aligns with recommended machine learning workflows that emphasise early data inspection to improve model reliability (Tongji University, n.d.).

Data preprocessing included handling inconsistent data types, removing non-informative variables, and preparing features for modelling. Feature selection was then performed to define predictor variables (X) and target variable (y).

The dataset was split into training and testing sets (80/20 split) with stratification to preserve class balance. A Decision Tree model was trained first as a baseline, followed by a tuned version to reduce overfitting. Additional models (Random Forest and Logistic Regression) were trained for benchmarking and comparison (Springer, 2017).

Model performance was evaluated using accuracy, precision, recall, F1-score, and confusion matrices.

# Dataset Overview

The dataset used in this study was obtained from Kaggle (Badole, 2024) and consists of medical measurements from breast tumour samples.

Each sample includes features such as:

- Clump thickness
- Cell size uniformity
- Cell shape uniformity
- Bare nuclei
- Bland chromatin
- Mitoses

The target variable indicates diagnosis:

- 2 = Benign
- 4 = Malignant

The dataset contains 698 records and 11 variables, making it suitable for binary classification tasks in supervised learning (Müller and Guido, 2016).

# Exploratory Data Analysis (EDA)

EDA was conducted to understand data structure, detect quality issues, and explore feature distributions. According to Müller and Guido (2016), EDA is essential for identifying preprocessing needs and improving model performance.

The dataset initially lacked column names and required manual assignment. Basic profiling included shape, data types, missing values, and statistical summaries.

Visual analysis included class distribution plots, histograms, boxplots, and correlation heatmaps. These revealed a mild class imbalance, correlated features, and variability in feature distributions.

A data type inconsistency was identified in the bare_nuclei feature and corrected during preprocessing. This highlights the importance of early data inspection as emphasised in machine learning literature (Tongji University, n.d.).

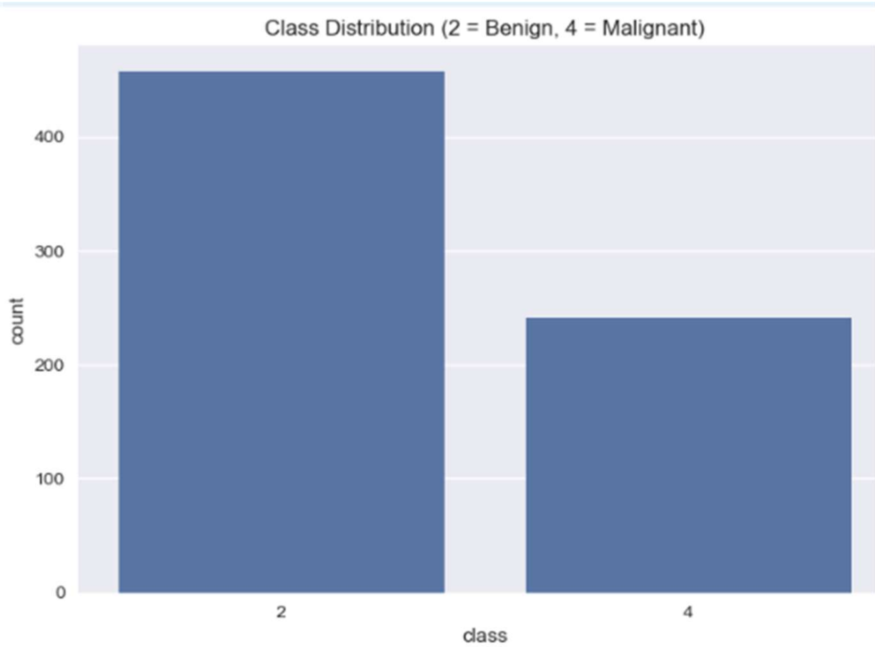
Overall, the dataset was found to be suitable for classification tasks due to its structured numerical features and clear class separation.

# Visual Analysis

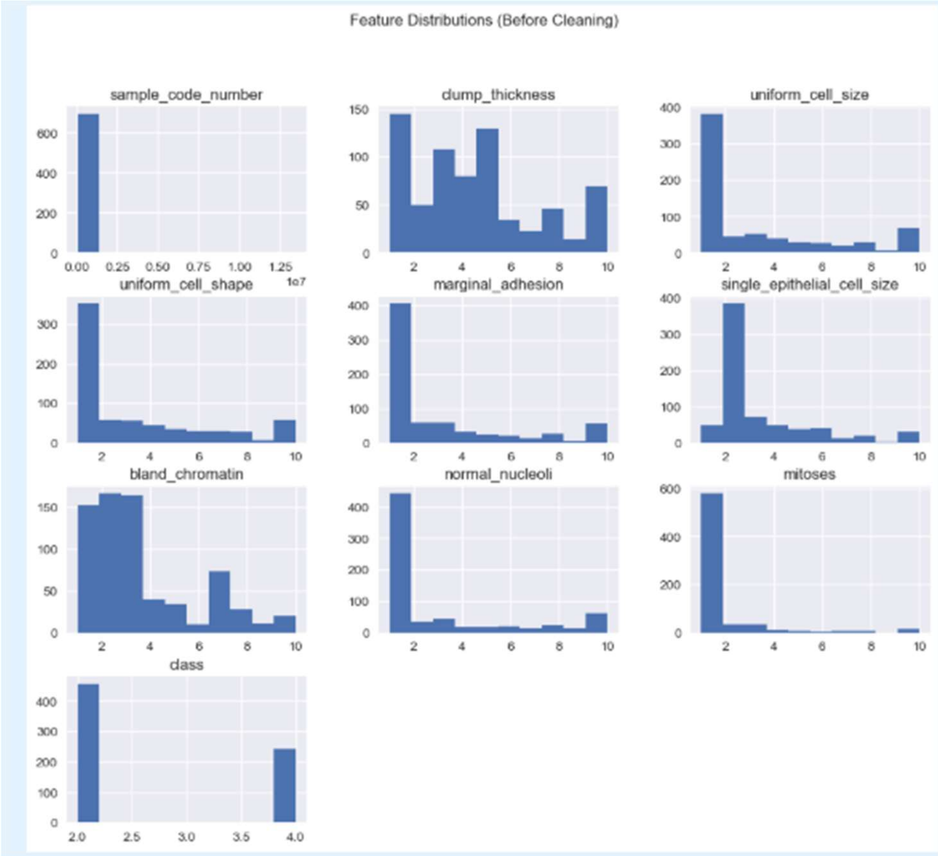
Visualisations were created to better understand the dataset before cleaning.

The following visualisations were produced:

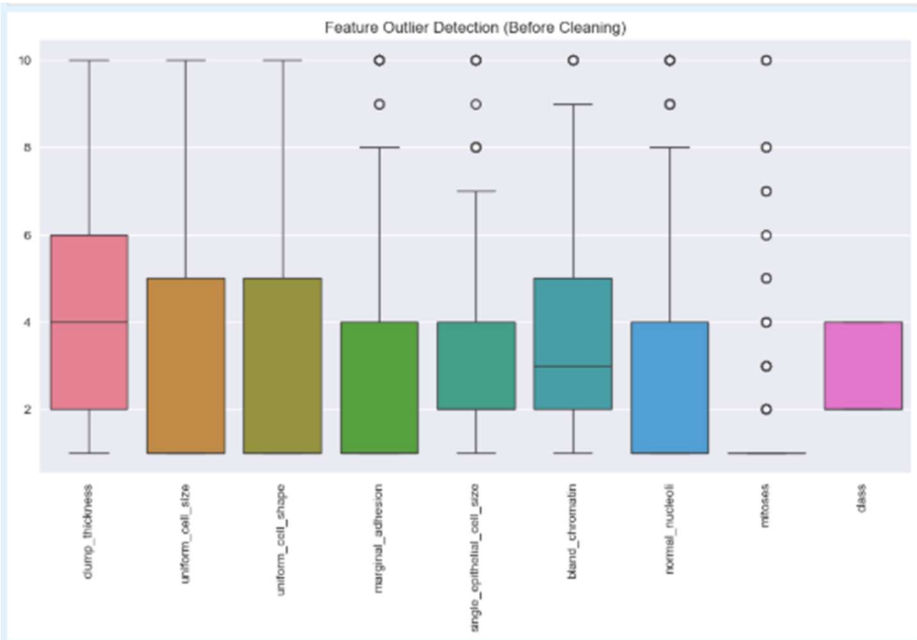
- Class distribution plot



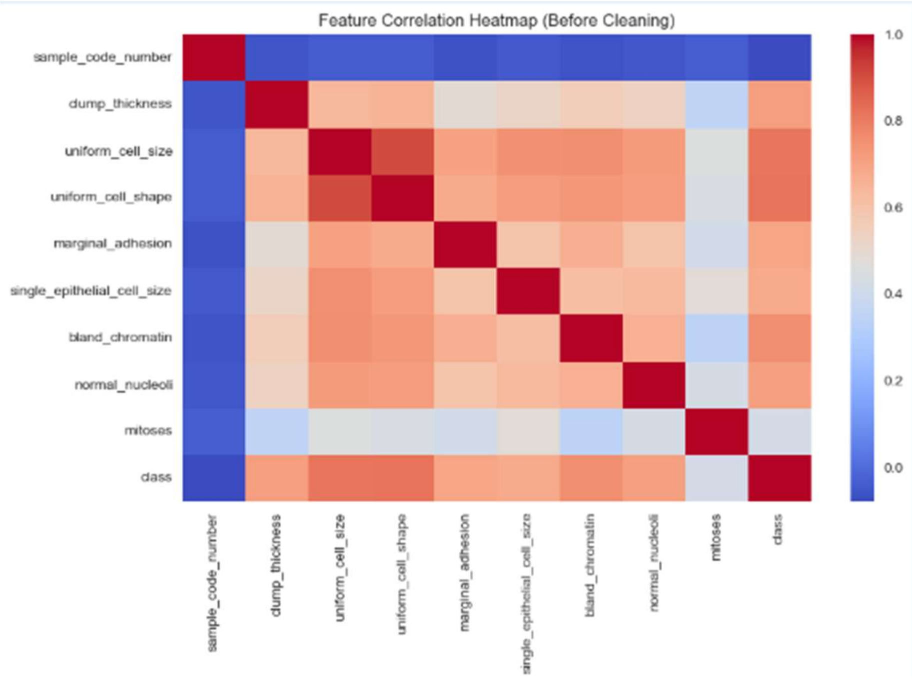

- Feature histograms




- Boxplots for outlier detection



- Correlation heatmap




The visual analysis revealed:

- Mild class imbalance between benign and malignant cases
- Some correlated variables
- Potential outliers in biological measurement features
- Strong relationships between several tumour characteristics

These findings justified further preprocessing and model training.

# Data Cleaning and Preprocessing

Data cleaning was performed to ensure compatibility with machine learning models.

The bare_nuclei column was corrected by converting invalid entries to missing values, then transforming the column into a numeric format and filling missing values with the median. This ensured consistency across all features.

The sample_code_number column was removed as it did not contribute to prediction. The target variable was encoded into binary form:

- 0 = Benign
- 1 = Malignant

After cleaning, the dataset contained no missing values and was fully numeric.

## Feature Engineering and Feature Selection

The target variable (y) was defined as the diagnosis class. All medically relevant features were used as predictors (X), while non-informative columns such as identifiers were removed.

This ensured that only clinically meaningful features were used for model training, consistent with standard machine learning practice (Müller and Guido, 2016).

## Train-Test Split

The dataset was split into 80% training and 20% testing data. Stratified sampling was used to maintain class distribution consistency across both sets.

A random state was applied to ensure reproducibility of results (Springer, 2017).

## Model Training and Evaluation

Three classification algorithms were trained and evaluated:

1. Decision Tree Classifier
2. Random Forest Classifier
3. Logistic Regression

The models were evaluated using:

- Accuracy
- Confusion matrix
- Precision
- Recall
- F1-score

## Decision Tree Model

The baseline Decision Tree achieved 93.6% accuracy but produced 8 false negatives, meaning some malignant cases were missed.

## Tuned Model

Hyperparameter tuning (max depth, minimum samples split, and leaf size) did not improve performance. Both models showed identical results, indicating limited benefit from tuning on this dataset.

This suggests that decision boundaries were already well-defined in the dataset (Müller and Guido, 2016).

## Random Forest Model

The Random Forest model improved performance by combining multiple decision trees to reduce variance and overfitting (Müller and Guido, 2016).

**Results:**

- Accuracy: 97.1%
- False Negatives: 4
- False Positives: 0

This model significantly improved cancer detection performance and reduced missed malignant cases.

## Logistic Regression Model

Logistic Regression was used as a linear benchmark model.

Results:

- Accuracy: 97.1%
- False Negatives: 4

This indicates that the dataset is nearly linearly separable, allowing simple models to perform well (Müller and Guido, 2016).

## Model Comparison

A comparison was conducted between all classification models.

| Model | Accuracy | False Negatives | Key Insight |
|---|---|---|---|
| Decision Tree | 93.6% | 8 | Simple but less reliable |
| Tuned Decision Tree | 93.6% | 8 | No improvement after tuning |
| Random Forest | 97.1% | 4 | Best overall performance |
| Logistic Regression | 97.1% | 4 | Strong linear benchmark |

## Key Findings

- Random Forest and Logistic Regression achieved the highest accuracy
- Decision Tree models produced more false negatives
- Reducing false negatives is critical in healthcare applications
- Ensemble models provided more reliable cancer detection

## Business Interpretation

The results demonstrate that machine learning can support medical schemes in improving decision-making efficiency and accuracy.

The models can help:

- Identify high-risk patients faster
- Reduce delays in benefit approval
- Improve consistency in cancer classification
- Support clinical decision-making

Random Forest is the most suitable model due to its lower false negative rate and higher reliability in detecting malignant cases (Springer, 2017).

In a real-world healthcare environment, such systems could be integrated into decision-support platforms to assist medical professionals in prioritising urgent patient cases.

# Conclusion

This project successfully applied supervised machine learning techniques to classify breast tumours as benign or malignant using the Breast Cancer Wisconsin Dataset.

The analysis followed a complete data analytics workflow including exploratory data analysis, preprocessing, feature engineering, model training, evaluation, retraining, and benchmarking.

All models performed well, with Random Forest and Logistic Regression achieving the highest accuracy of 97.1%.

The Decision Tree model provided useful interpretability but produced more false negatives, making it less suitable for medical deployment.

Overall, the Random Forest classifier was identified as the best-performing model due to its strong predictive accuracy, lower false negative rate, and better generalisation capability.

The project demonstrates how machine learning can support healthcare organisations in improving early cancer detection and accelerating decision-making processes for medical benefit approvals.

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